Results with BCS theory

Now that we can solve Eqs. **??** and **??** let's see the results for some different materials and various temperatures. In particular i have tested Nb, Pb and Al.

For different materials we need the experimental values of thecritical temperature T_c, under which they become superconductors, and the zero temperature value of the enrgy gap Δ_0. Actually for Δ_0 we extract the value from the BCS relation

$$\frac{2\Delta_0}{K_B T_c} = \alpha, \tag{1}$$

where α is ~ 3.53 for weak-coupling materials, that is the limit where BCS theory and MB formulas holds, and is greater for strong-coupling materials such as Nb and Pb. In Tab. 1 below the T_c and α values are showed.

Since $\Delta = \Delta(T)$ is a function of the temperature and is zero at $T \geq T_c$ i have extrapolate it in the BCS approximation as

$$\Delta = \Delta_0 \ \tanh\left(e^{\gamma}\sqrt{\frac{8}{7\xi(3)}}\ \sqrt{\frac{T_c}{T}-1}\right) \simeq \Delta_0 \ \tanh\left(1.74\ \sqrt{\frac{T_c}{T}-1}\right). \tag{2}$$

In the following we see some test for Nb, Pb and Al at different temperatures. In Fig. 1 i compared the real and imaginary conductivity for Nb, Pb and Al at $T = 0.8T_c$. In Fig.2 we see conductivity for different temperatures in Nb. In the last Fig. 3 i have plotted the conductivity of Niobium with the energy scale normalized by $2\Delta_0$ to check if the real part of σ/σ_n correctly drops down at $\hbar\omega = 2\Delta_0$ when $T = 0$, and if increasing T the gap moves to lower energies/frequencies.

Table 1: Superconducting materials experimental parameters.

Material	T_c(K)	α
Nb	9.3	3.65
Pb	7.2	3.95
Al	1.2	3.53

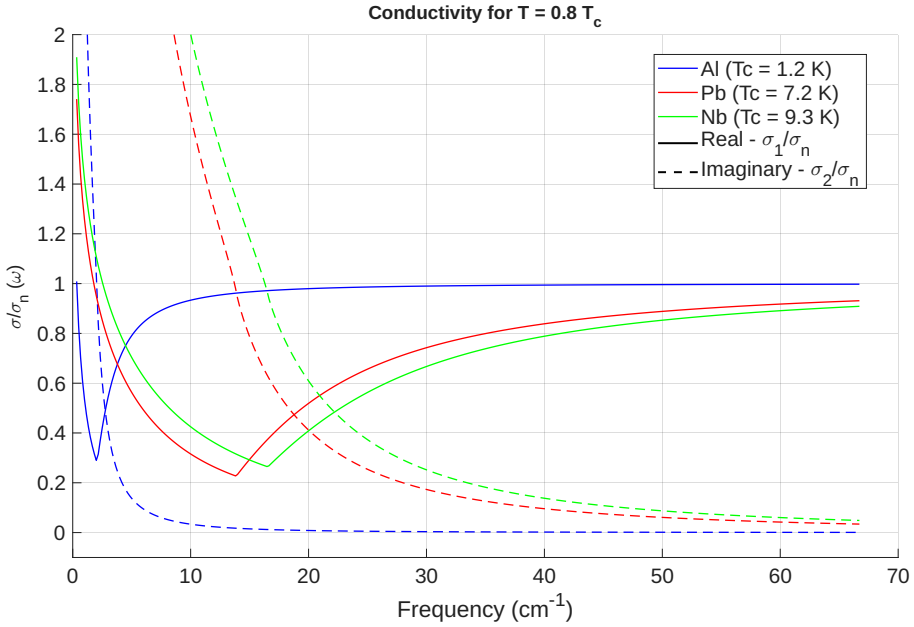


Figure 1: Complex conductivity for different materials.

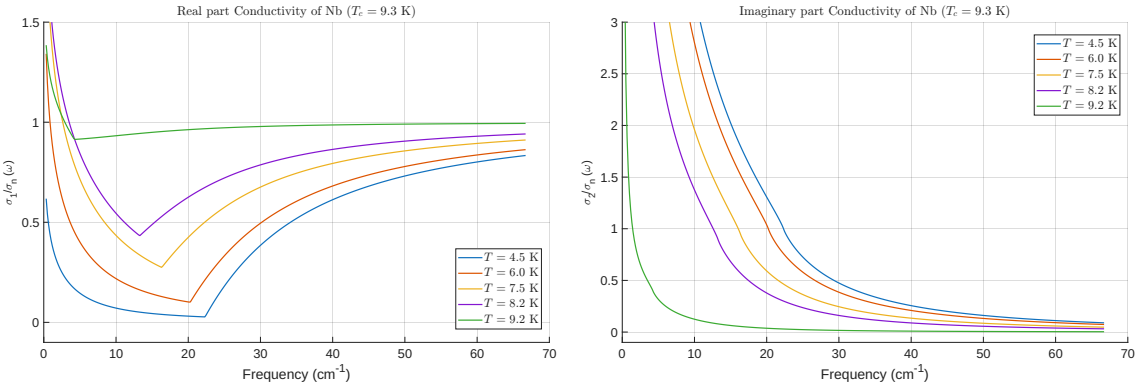


Figure 2: Real (sx) and imaginary (dx) conductivity for Nb.

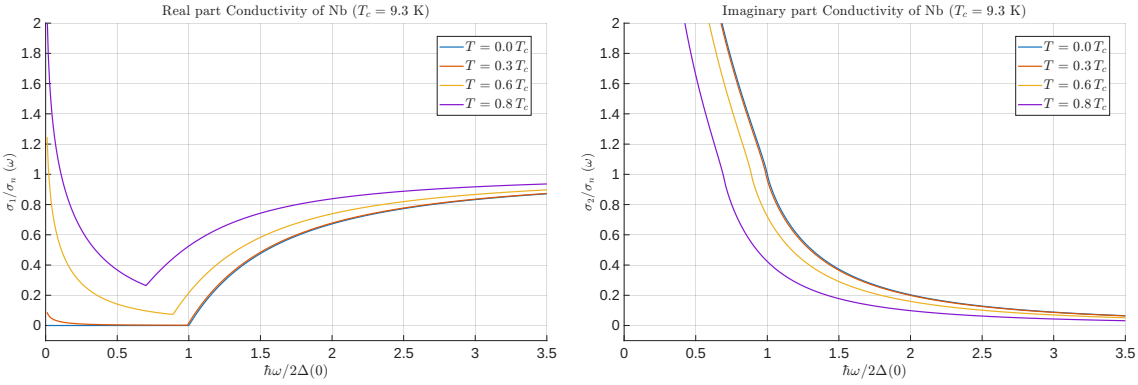


Figure 3: Real (sx) and imaginary (dx) conductivity for Nb in normalized units, to check the gap corresponds at $T = 0$.

Fit with experimental data

To evaluate the performances of the code that i've written, i compare the numerical solutions with some experimental datas extracted from [ref]. The results are showed in the Fig.4 below. We can see that the fit is pretty bad, expecially for lower T, but we espected

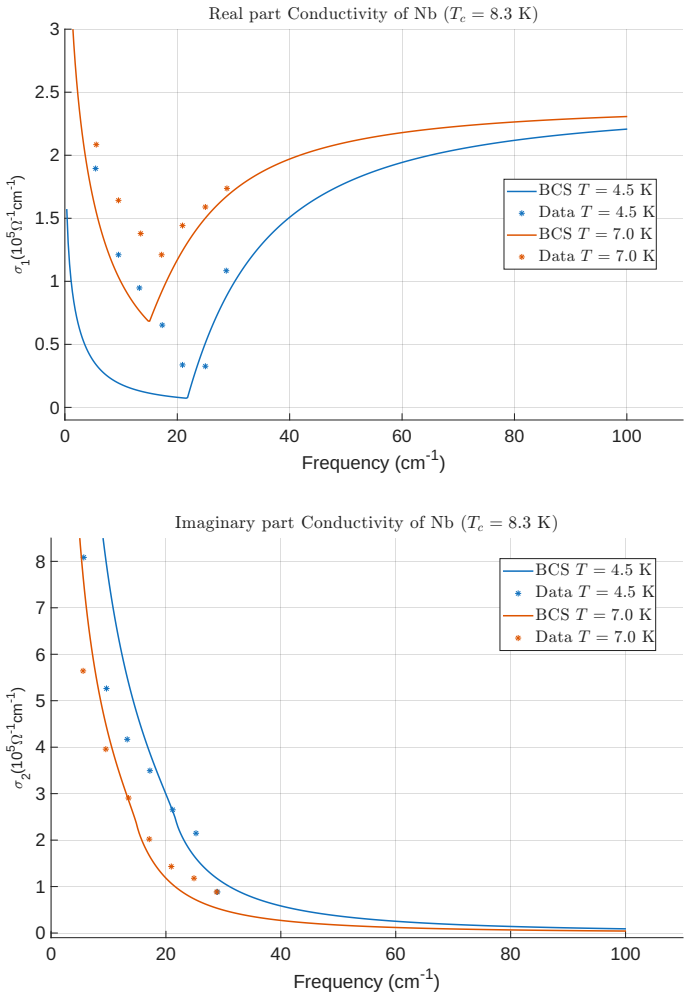


Figure 4: Real (sx) and imaginary (dx) conductivity for Nb compared with experimental data.

this for Niobium because is a strong-coupling superconducting material while BCS is a weak-coupling theory ($\alpha = 3.65$).